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SE08_02

Jurnal 13

OUTPUT: SCREENSHOOT LAYAR HASIL RUNNING QUERY MONGOSHHELL

Disediakan dataset berikut:

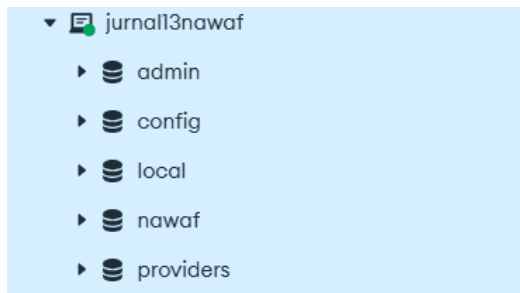
```
db.toko.insertMany([
  {
    "name" : "Toko Berkah",
    "category" : "Elektronik",
    "city" : "Jakarta",
    "address" : { "street" : "Jl. Sudirman", "zipcode" : "10220" },
    "products" : [
      { "item" : "TV", "price" : 3500000, "stock" : 10 },
      { "item" : "Kulkas", "price" : 2800000, "stock" : 5 }
    ]
  },
  {
    "name" : "Toko Maju",
    "category" : "Elektronik",
    "city" : "Bandung",
    "address" : { "street" : "Jl. Asia Afrika", "zipcode" : "40111" },
    "products" : [
      { "item" : "HP", "price" : 1500000, "stock" : 20 },
      { "item" : "Laptop", "price" : 8000000, "stock" : 3 }
    ]
  },
  {
    "name" : "Toko Salim Bakery",
    "category" : "Elektronik",
    "city" : "Jakarta",
    "address" : { "street" : "Jl. Sudirman", "zipcode" : "10220" },
    "products" : [
      { "item" : "Monitor LED", "price" : 7700000, "stock" : 11 },
```

```

    { "item" : "Airfryer Panasonic", "price" : 4850000, "stock" : 7 }
  ]
},
{
  "name" : "Toko Sudinotech",
  "category" : "Elektronik",
  "city" : "Surabaya",
  "address" : { "street" : "Jl. Sudirman", "zipcode" : "10220" },
  "products" : [
    { "item" : "RTX 55000", "price" : 8500000, "stock" : 20 },
    { "item" : "Set PC Gaming Radeon", "price" : 12000000, "stock" : 9 }
  ]
},
])

```

Jawaban



```

  })
  < {
    acknowledged: true,
    insertedIds: {
      '0': ObjectId('6a112a251be0b8c9e678d730'),
      '1': ObjectId('6a112a251be0b8c9e678d731'),
      '2': ObjectId('6a112a251be0b8c9e678d732'),
      '3': ObjectId('6a112a251be0b8c9e678d733')
    }
  }
}

```

1. Tambahkan 1 toko baru bebas ke koleksi toko, dengan ketentuan:

- Minimal memiliki 2 produk di dalam array products
- Isi semua field: name, category, city, address, products Hint:

```
> db.toko.insertOne({
  "name": "Toko Cahaya",
  "category": "Fashion",
  "city": "Surabaya",
  "address": { "street": "Jl. Tunjungan", "zipcode": "60275" },
  "products": [
    { "item": "Kemeja Batik", "price": 250000, "stock": 15 },
    { "item": "Celana Chino", "price": 350000, "stock": 12 }
  ]
})
< {
  acknowledged: true,
  insertedId: ObjectId('6a112ab31be0b8c9e678d734')
}
test>
```

2. Tampilkan semua toko yang berada di kota Jakarta

```
> db.toko.find({ "city": "Jakarta" })
< {
  _id: ObjectId('6a112a251be0b8c9e678d730'),
  name: 'Toko Berkah',
  category: 'Elektronik',
  city: 'Jakarta',
  address: {
    street: 'Jl. Sudirman',
    zipcode: '10220'
  },
  products: [
    {
      item: 'TV',
      price: 3500000,
      stock: 10
    },
    {
      item: 'Kulkas',
      price: 2800000,
      stock: 5
    }
  ]
}
{
  _id: ObjectId('6a112a251be0b8c9e678d732'),
```

3. Tampilkan toko yang memiliki produk dengan price lebih dari 3.000.000.

Hint:

- Gunakan dot notation: products.price
- Gunakan operator: \$gt

```
> db.toko.find({ "products.price": { $gt: 3000000 } })
< {
  _id: ObjectId('6a112a251be0b8c9e678d730'),
  name: 'Toko Berkah',
  category: 'Elektronik',
  city: 'Jakarta',
  address: {
    street: 'Jl. Sudirman',
    zipcode: '10220'
  },
  products: [
    {
      item: 'TV',
      price: 3500000,
      stock: 10
    },
    {
      item: 'Kulkas',
      price: 2800000,
      stock: 5
    }
  ]
}
```

4. Ubah category Toko Berkah dari Fashion menjadi Aksesoris.

Tambahkan 1 produk baru ke dalam array products milik Toko Maju menggunakan \$push.

Hint:

- Filter berdasarkan name: Toko Maju
- Gunakan operator \$push untuk menambah elemen ke array

```
> db.toko.updateOne(
  { "name": "Toko Berkah" },
  { $set: { "category": "Aksesoris" } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
> db.toko.updateOne(
  { "name": "Toko Maju" },
  { $push: { "products": { "item": "Mouse Wireless", "price": 250000, "stock": 15 } } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
test>
```

5. Hitung jumlah toko per city, urutkan dari kota yang paling banyak tokonya.

Gunakan:

- \$group dengan \$sum: 1
- \$sort descending (-1)

```
> db.toko.aggregate([
  { $group: { _id: "$city", totalToko: { $sum: 1 } } },
  { $sort: { totalToko: -1 } }
])
< {
  _id: 'Surabaya',
  totalToko: 2
}
{
  _id: 'Jakarta',
  totalToko: 2
}
{
  _id: 'Bandung',
  totalToko: 1
}
test>
```

6. Hitung rata-rata harga produk per category.

Tahapan pipeline:

- \$unwind: '\$products' (bongkar array products dulu)
- \$group: kelompokkan per category, hitung \$avg dari products.price
- \$sort: urutkan sesuai kebutuhan

```
> db.toko.aggregate([
  { $unwind: "$products" },
  { $group: { _id: "$category", rataRataHarga: { $avg: "$products.price" } } },
  { $sort: { rataRataHarga: -1 } }
])
< {
  _id: 'Elektronik',
  rataRataHarga: 6114285.714285715
}
{
  _id: 'Aksesoris',
  rataRataHarga: 3150000
}
{
  _id: 'Fashion',
  rataRataHarga: 300000
}
test>
```

7. Tambahkan 1 produk baru ke dalam array products milik Toko Maju menggunakan \$push.

Hint:

- Filter berdasarkan name: Toko Maju
- Gunakan operator \$push untuk menambah elemen ke array

```
> db.toko.updateOne(
  { "name": "Toko Maju" },
  { $push: { "products": { "item": "Keyboard Mechanical", "price": 750000, "stock": 8 } } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
test>
```

8. Tampilkan harga produk tertinggi (\$max) per city.

Hint:

- Unwind products terlebih dahulu
- Gunakan \$max pada products.price

```
> db.toko.aggregate([
  { $unwind: "$products" },
  { $group: { _id: "$city", hargaTertinggi: { $max: "$products.price" } } }
])
< {
  _id: 'Surabaya',
  hargaTertinggi: 12000000
}
{
  _id: 'Jakarta',
  hargaTertinggi: 7700000
}
{
  _id: 'Bandung',
  hargaTertinggi: 8000000
}
test> |
```

9. Buat pipeline lengkap dengan 4 tahap:

- \$match — filter hanya toko category Elektronik
- \$unwind — bongkar array products
- \$group — hitung total stock per city
- \$sort — urutkan dari total stock terbesar

```
> db.toko.aggregate([
  { $match: { "category": "Elektronik" } },
  { $unwind: "$products" },
  { $group: { _id: "$city", totalStock: { $sum: "$products.stock" } } },
  { $sort: { totalStock: -1 } }
])
< {
  _id: 'Bandung',
  totalStock: 46
}
{
  _id: 'Surabaya',
  totalStock: 29
}
{
  _id: 'Jakarta',
  totalStock: 18
}
test> |
```

10. Hapus toko yang memenuhi dua kondisi sekaligus:

- category adalah Fashion
- city adalah Surabaya

Hint: Gunakan \$and atau tulis kedua kondisi dalam satu objek filter

```
> db.toko.deleteOne({
  $and: [
    { "category": "Fashion" },
    { "city": "Surabaya" }
  ]
})
< {
  acknowledged: true,
  deletedCount: 1
}
test>
```